Resonant Decay of Kinetic Alfvén Waves and Implication on Spectral Cascading

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Outline

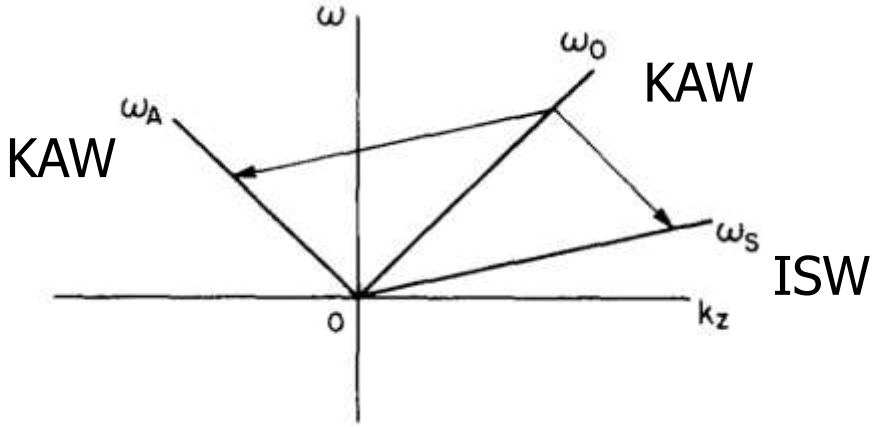
- Background and Motivation
- Nonlinear Gyrokinetic Equations
- Nonlinear Mode Equation of KAW Turbulence
- Parametric Decay and Spectral Cascades
- Summary and Discussion

Background

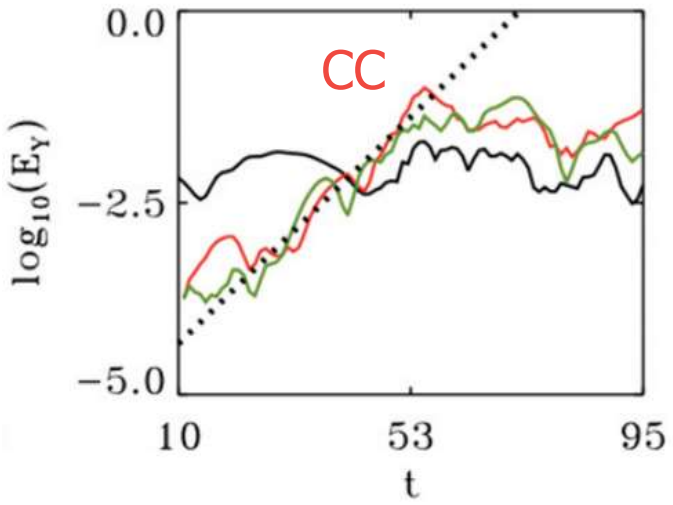
- Shear Alfvén waves (SAWs): incompressible electromagnetic oscillations
  - solar wind, solar corona, interstellar medium, tokamaks
- Kinetic Alfvén waves (KAWs) with FLR correction generated by:
  - phase-mixing due to plasma nonuniformity [Chen&Hasegawa PoF74]
  - wave-particle/EP interaction [Fasoli et al. NF07]
  - nonlinear mode-coupling [Chen&Zonca PoP13] $\Longrightarrow$ multiple-mode interactions
- Low-frequency Alfvénic turbulence [Chen&Zonca RMP16; Schekochihin et al. ApJS09]
  - burning plasmas: heating, turbulent saturation, anomalous transport
  - space plasmas: stationary energy spectra, particle heating and acceleration

Motivation

- Nonlinear wave-wave interactions ⇐ weak turbulence theory, to give quantitative description of collisionless transport and turbulent evolution
  - parametric decay: KAW ⟶ ion sound wave (ISW) + KAW [Chen&Zonca EPL11]
  - modulational instability: KAW ⟶ convective cell (CC) [Zonca et al. EPL15]
- Parametric decay KAW ⟶ KAW+KAW investigated here



$$(\omega_0, \boldsymbol{k}_0) \longrightarrow (\omega_A, \boldsymbol{k}_A) + (\omega_S, \boldsymbol{k}_S)$$



Nonlinear Gyrokinetic Equations

- Assumptions: $\omega_k \ll \Omega_i$, $k_\parallel \ll k_\perp$, low-β uniform plasmas, $\boldsymbol{B}_0 = const$
- Perturbed distribution functions (Maxwellian background):

$$\delta f_j = -(q/T)_j F_{Mj} \delta\phi + e^{-\boldsymbol{\rho}_j \cdot \boldsymbol{\nabla}} \delta g_j$$

  - multiple-modes co-exist $\Longrightarrow$ $\delta\phi = \sum_{\boldsymbol{k}} \left[\delta\phi_k(t) e^{i\boldsymbol{k}\cdot\boldsymbol{r}} + c.c.\right]/2$

- Nonlinear gyrokinetic equation gives nonadiabatic responses δg_j

$$\left[\partial_t + v_\parallel \hat{\boldsymbol{b}} \cdot \boldsymbol{\nabla} + (c/B_0)\hat{\boldsymbol{b}} \times \boldsymbol{\nabla} \langle \delta L_{g,j} \rangle_\alpha \cdot \boldsymbol{\nabla}\right] \delta g_j = (q/T)_j F_{Mj} \partial_t \langle \delta L_{g,j} \rangle_\alpha$$

  - perpendicular convective nonlinearity
  - $\langle \delta L_{g,j} \rangle_\alpha = J_k(k_\perp \rho_i)(\delta\phi - v_\parallel \delta A_\parallel / c)$, $J_k(k_\perp \rho_i) = \langle \exp(\boldsymbol{\rho}_i \cdot \boldsymbol{\nabla}) \rangle_\alpha$

Nonlinear Gyrokinetic Equations

- Quasi-neutrality condition

$$(1+\tau)\delta\phi_k = T_e/(n_0 e)\langle J_k \delta g_{k,i} - \delta g_{k,e}\rangle_v$$

- Nonlinear gyrokinetic vorticity equation [Chen&Hasegawa JGR91]

$$ik_{\|}\delta J_{\|k} + (n_0 e^2/T_i)(1-\Gamma_k)\partial_t \delta\phi_k$$

$$= \sum_{\boldsymbol{k}=\boldsymbol{k}'+\boldsymbol{k}''} \Lambda^{k'}_{k''}[\delta A_{\|k'}\delta J_{\|k''}/c - e\langle (J_k J_{k'} - J_{k''})\delta L_{k'}\delta g_{k'',i}\rangle_v]$$

  - derived from parallel Ampere's law, QN condition, and NLGKE
  - LHS: field line bending, plasma inertia
  - RHS: Maxwell stress (MX, $\delta\boldsymbol{J}_{\|}\times\delta\boldsymbol{B}_{\perp}$), Reynolds stress (RS, $\delta\boldsymbol{v}_{\perp}\cdot\nabla\delta\boldsymbol{v}_{\perp}$)
  - polarization nonlinearity neglected in NLGKV for $|k_{\perp}\rho_i|^2/|\omega_k/\Omega_i| \gtrsim \mathcal{O}(1)$

Linear Physics of KAWs

- Field variables: $\delta\phi_k$ and $\delta\psi_k = \omega_k \delta A_{\parallel k}/(ck_\parallel)$
- Linear responses derived from GKE assuming $v_{te} \gg |\omega_k/k_\parallel| \gg v_{ti}$
- KAW linear dispersion relation

$$\varepsilon_{Ak} = \frac{1-\Gamma_k}{b_k} - \frac{k_\parallel^2 v_A^2}{\omega_k^2}\sigma_k \quad \Longrightarrow \quad \omega_k^2 = k_\parallel^2 v_A^2 \frac{\sigma_k b_k}{1-\Gamma_k}$$

- Linear property of KAWs $\delta\psi_k = \sigma_k \delta\phi_k$, $\sigma_k = 1 + \tau(1-\Gamma_k)$
  - $\sigma_k \neq 1 \Longrightarrow$ ideal-MHD constraint breaks down
  - finite $\delta E_\parallel \Longrightarrow$ particle heating and acceleration
  - finite perpendicular group velocity $v_{g\perp}$

Nonlinear Mode Equation

- The nonlinear electron response derived from NLGKE:

$$\delta g^{(2)}_{k,e} \simeq \sum_{\boldsymbol{k}=\boldsymbol{k}'+\boldsymbol{k}''} (i\Lambda^{k'}_{k''}/2k_\parallel)(eF_{Me}/T_e)(k'_\parallel/\omega_{k'} - k''_\parallel/\omega_{k''})\delta\psi_{k'}\delta\psi_{k''}$$

  - while the nonlinear ion response is negligible due to low-β assumption

- Nonlinear gyrokinetic vorticity equation

$$b_k\left(\frac{1-\Gamma_k}{b_k}\delta\phi_k - \frac{k_\parallel^2 v_A^2}{\omega_k^2}\delta\psi_k\right) = -\sum_{\boldsymbol{k}=\boldsymbol{k}'+\boldsymbol{k}''} (i\Lambda^{k'}_{k''}/2\omega_k)[(\Gamma_{k'}-\Gamma_{k''})\delta\phi_{k'}\delta\phi_{k''}$$
$$+(k'_\parallel k''_\parallel/\omega_{k'}\omega_{k''})(b_{k'}-b_{k''})v_A^2\delta\psi_{k'}\delta\psi_{k''}]$$

RS + MX

- Quasi-neutrality condition from nonlinear particle responses

$$\delta\psi_k = \sigma_k\delta\phi_k + \sum_{\boldsymbol{k}=\boldsymbol{k}'+\boldsymbol{k}''} (i\Lambda^{k'}_{k''}/2k_\parallel)(k'_\parallel/\omega_{k'} - k''_\parallel/\omega_{k''})\delta\psi_{k'}\delta\psi_{k''}$$

NL $\delta E_\parallel$

Nonlinear Mode Equation

- Nonlinear mode equation for multiple-mode interactions

$$b_k \varepsilon_{Ak} \delta\phi_k = -\sum_{\boldsymbol{k}=\boldsymbol{k}'+\boldsymbol{k}''} (i\Lambda^{k'}_{k''}/2\omega_k)\beta_{k',k''}\delta\phi_{k'}\delta\phi_{k''} \qquad (1)$$

  - nonlinear dynamics of multiple KAWs and strong/weak turbulent evolution

- Nonlinear coupling coefficient

$$\beta_{k',k''} = \sigma_{k'}\sigma_{k''}\left(\frac{s_{k''}}{\alpha_{k''}} - \frac{s_{k'}}{\alpha_{k'}}\right)\left(b_k\frac{s_k}{\alpha_k} + b_{k'}\frac{s_{k'}}{\alpha_{k'}} + b_{k''}\frac{s_{k''}}{\alpha_{k''}}\right) \qquad (2)$$

  - $\beta_{k',k''}$ contribution from MX + RS + NL $\delta E_\parallel$
  - $s_k \equiv \mathrm{sign}(\omega_k/k_\parallel) = \pm 1$: propagating direction along $\boldsymbol{B}_0$
  - $\alpha_k \equiv \left|\frac{\omega_k}{k_\parallel v_A}\right| = \sqrt{\frac{\sigma_k b_k}{1-\Gamma_k}}$: **monotonically increase with $|\boldsymbol{k}_\perp|$**
  - cascade behaviors in $k_\perp$ space $\Leftarrow$ dual-cascade for e-DWs [Hasegawa et al. PoF79]

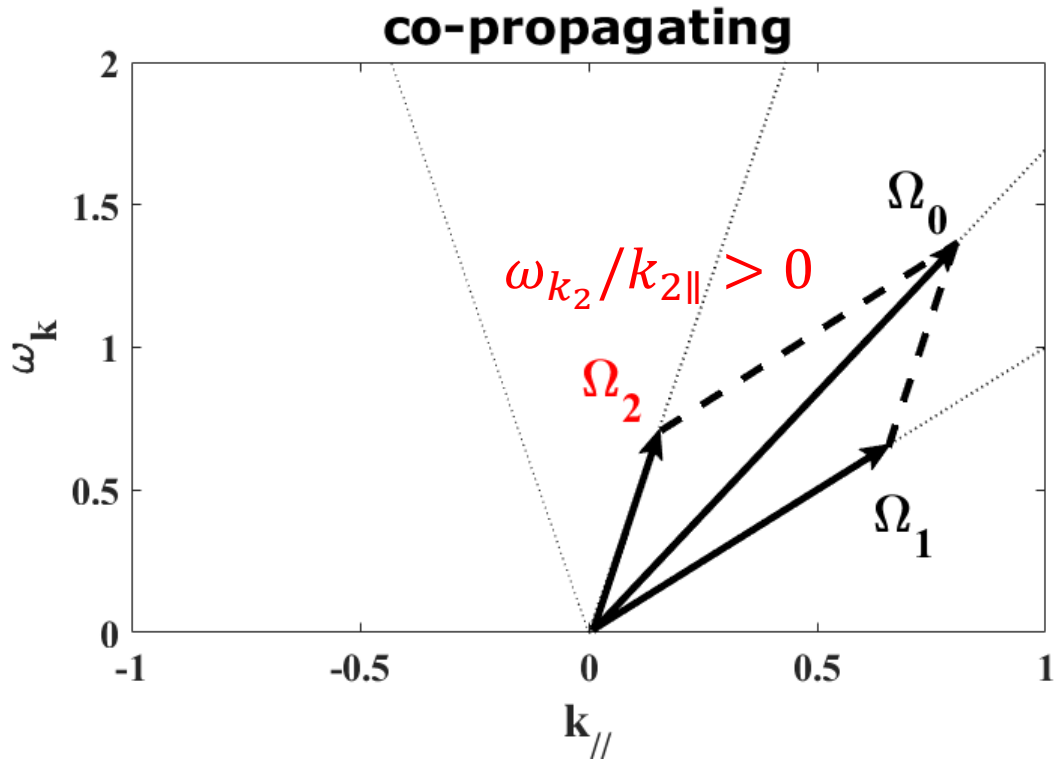
Parametric Decay

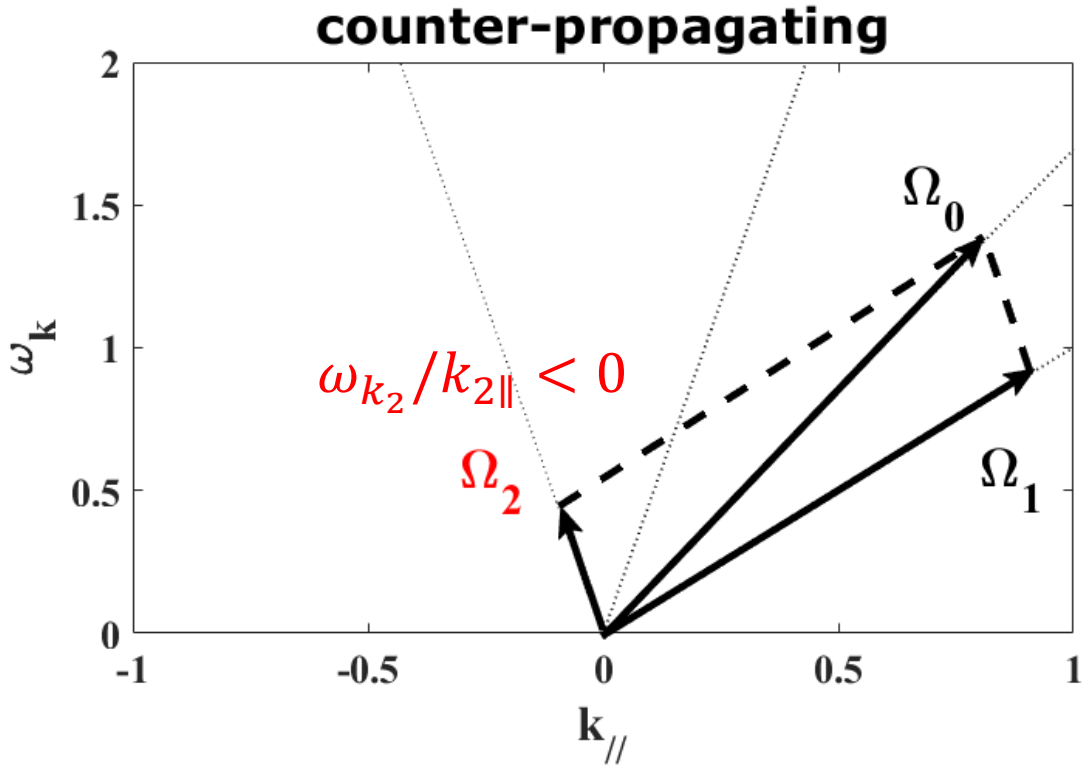
- Spontaneous decay: $\boldsymbol{\Omega_0 \longrightarrow \Omega_1 + \Omega_2}$

  - matching conditions: $\omega_{k_0} = \omega_{k_1} + \omega_{k_2}, \quad \boldsymbol{k_0 = k_1 + k_2}$
  - $\boldsymbol{\Omega_2}$ co-/counter-propagates with respect to $\boldsymbol{\Omega_0}$ and $\boldsymbol{\Omega_1}$

$$s_k \equiv \mathrm{sign}(\omega_k / k_\parallel) = \pm 1$$

$$\alpha_k \equiv \left| \frac{\omega_k}{k_\parallel v_A} \right| = \sqrt{\frac{\sigma_k b_k}{1 - \Gamma_k}}$$





Parametric Decay

- Coupled equations derived from nonlinear mode equation
  - taking $\delta\phi_k(t) = \Phi_k(t)e^{-i\omega_k t}$ and assuming $|\Phi_{k_0}| \gg |\Phi_{k_1}|, |\Phi_{k_2}|$

$$b_{k_1}\varepsilon_{Ak1}\Phi_{k_1} = -i(\Lambda_{k_2}^{k_1}/2\omega_{k_1})\beta_{k_2,k_0}\Phi_{k_2}^{*}\Phi_{k_0}$$

$$b_{k_2}\varepsilon_{Ak2}\Phi_{k_2} = -i(\Lambda_{k_2}^{k_1}/2\omega_{k_2})\beta_{k_0,k_1}\Phi_{k_0}\Phi_{k_1}^{*}$$

- Parametric dispersion relation for 3-KAW spontaneous decay

$$(\partial_t - \gamma_{dk_1})(\partial_t - \gamma_{dk_2}) = \frac{(\Lambda_{k_2}^{k_1})^2\beta_{k_2,k_0}\beta_{k_0,k_1}}{16(1-\Gamma_{k_1})(1-\Gamma_{k_2})}|\Phi_{k_0}|^2 \equiv \gamma_{ND}^2 \qquad (3)$$

  - squared nonlinear drive $\gamma_{ND}^2 > 0 \Longrightarrow$ determine cascade behaviors

Spectral Cascades

- Intensity profiles of γ_{ND}^2/Ω_i^2 for $\boldsymbol{\Omega}_2$ being co-/counter-propagating

$$\gamma_{ND}^2 = \frac{(\Lambda_{k_2}^{k_1})^2 \beta_{k_2,k_0}\beta_{k_0,k_1}}{16(1-\Gamma_{k_1})(1-\Gamma_{k_2})}|\Phi_{k_0}|^2 \propto (s_{k_0}\alpha_{k_2} - s_{k_2}\alpha_{k_0})(s_{k_1}\alpha_{k_0} - s_{k_0}\alpha_{k_1})$$

$$\alpha_k \equiv \left|\frac{\omega_k}{k_\parallel v_A}\right| = \sqrt{\frac{\sigma_k b_k}{1-\Gamma_k}}$$

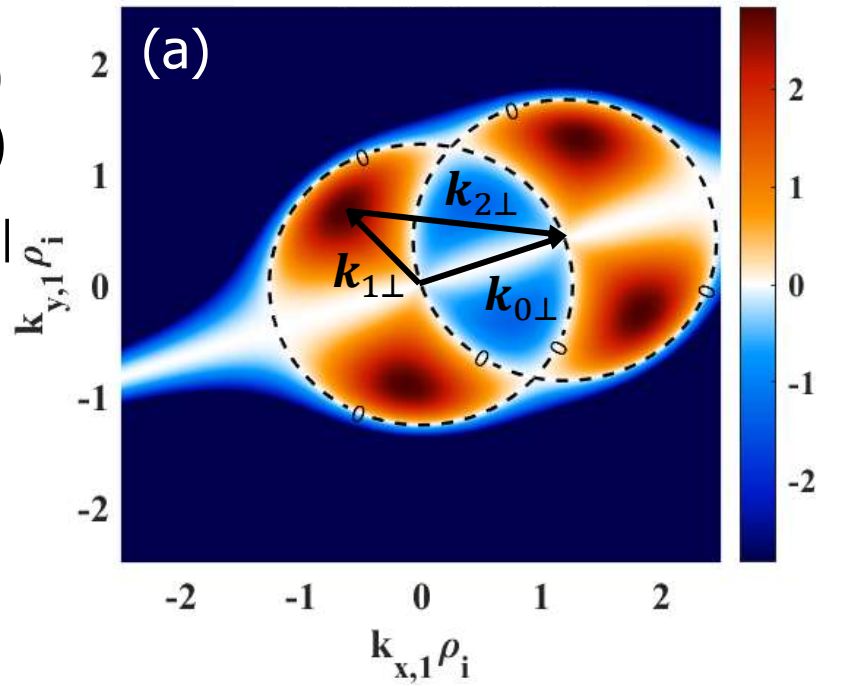
  - $\boldsymbol{k}_{0\perp}$ fixed, $\boldsymbol{k}_{1\perp}$ varies, and $\boldsymbol{k}_{2\perp} = \boldsymbol{k}_{0\perp} - \boldsymbol{k}_{1\perp}$

$\gamma_{ND}^2 \propto (\alpha_{k_2} - \alpha_{k_0}) \times (\alpha_{k_0} - \alpha_{k_1})$

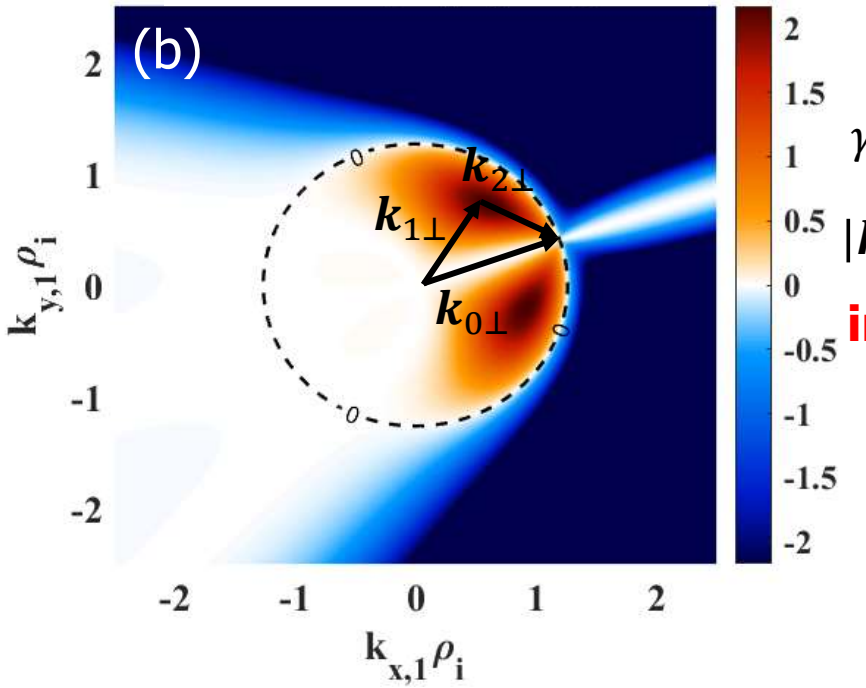
$|\boldsymbol{k}_{2\perp}| > |\boldsymbol{k}_{0\perp}| > |\boldsymbol{k}_{1\perp}|$

dual-cascade

co-propagating



counter-propagating



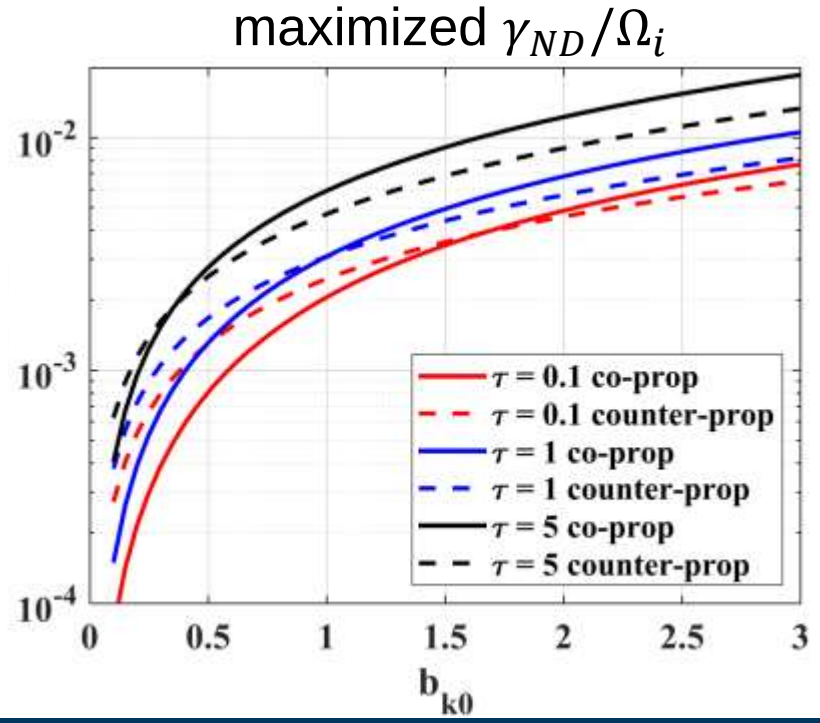
$\gamma_{ND}^2 \propto (\alpha_{k_0} - \alpha_{k_1})$

$|\boldsymbol{k}_{0\perp}| > |\boldsymbol{k}_{1\perp}|, |\boldsymbol{k}_{2\perp}|$

inverse-cascade

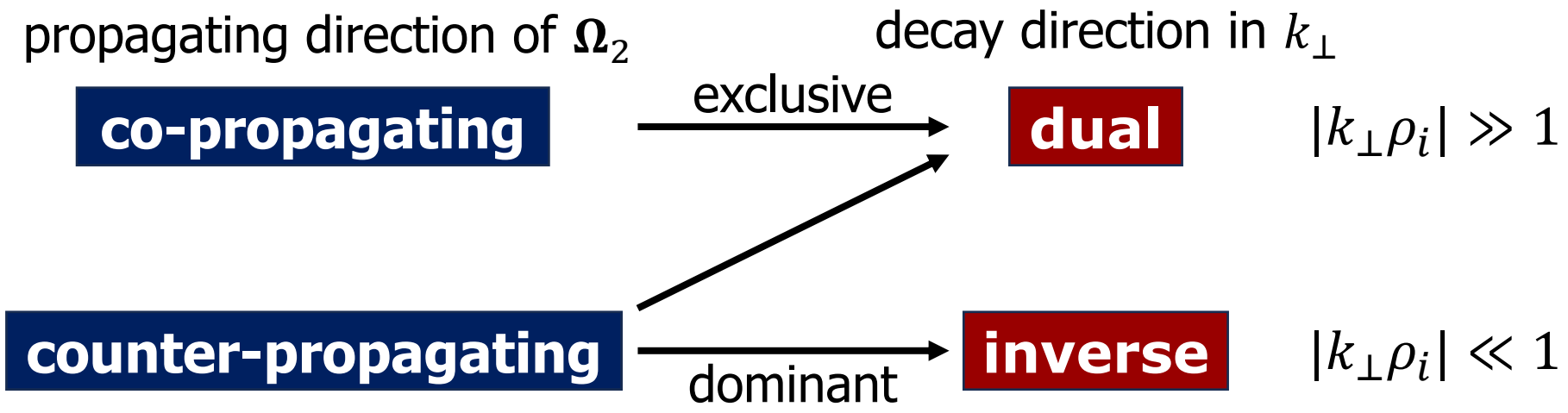
Spectral Cascades

- Parametric dependence of maximized γ_{ND} on b_{k_0} and τ
  - maximized γ_{ND} increases with b_{k_0} monotonically
  - small b_{k_0}: counter-propagating dominant (dashed lines)
  - large b_{k_0}: co-propagating dominant (solid lines)



Summary

- Nonlinear mode equation of KAW turbulence derived and three-KAW resonant decay analyzed
- Spectral cascades in $k_\perp$ space of parametric decay process:



Discussion

- Reduced equations in the $b_k \lesssim \mathcal{O}(1)$ limit similar to Hasegawa-Mima equation
  - counter-propagating KAWs:

$$\nabla_\perp \cdot [\partial_t^2 - v_A^2 \nabla_\parallel^2 + (3/4+\tau) v_A^2 \rho_i^2 \nabla_\parallel^2 \nabla_\perp^2] \nabla_\perp \delta\phi + (2c/B_0) \partial_t [(\hat{b} \cdot \nabla_\perp \delta\phi \times \nabla_\perp) \nabla_\perp^2 \delta\phi] = 0$$

  - co-propagating KAWs:

$$\nabla_\perp \cdot [\partial_t^2 - v_A^2 \nabla_\parallel^2 + (3/4+\tau) v_A^2 \rho_i^2 \nabla_\parallel^2 \nabla_\perp^2] \nabla_\perp \delta\phi + (3/4+\tau)(c\rho_i^2/B_0) \partial_t [(\hat{b} \cdot \nabla_\perp \delta\phi \times \nabla_\perp) \nabla_\perp^4 \delta\phi] = 0$$

- Future work along this line: stationary spectra of KAW turbulence; cascade directions and spectra of different quadratic invariants; more detailed numerical simulation

Thank you for your attention!

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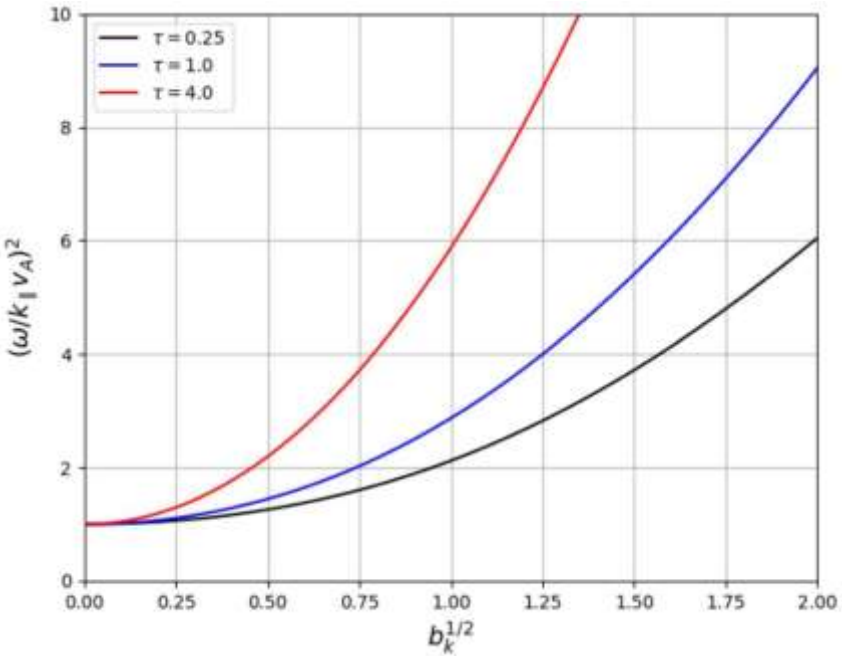
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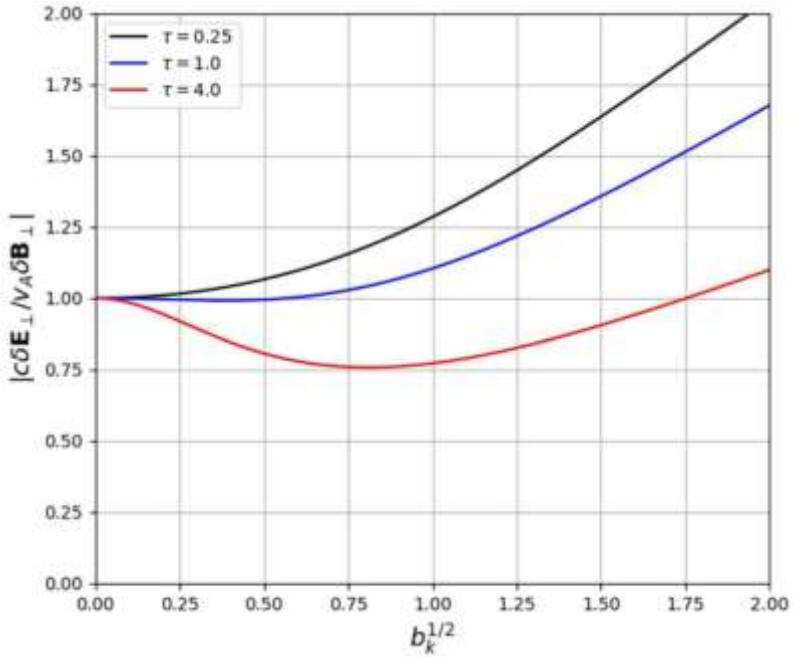
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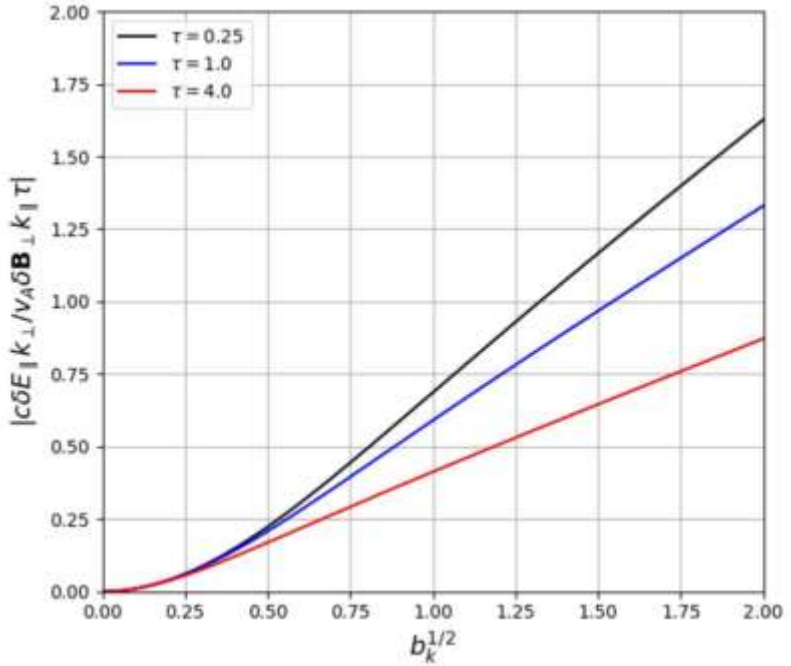
K.Shen, Z. Cheng, and Z. Qiu, Phys. Plasmas 31, 042309 (2024)



$(\omega_{\boldsymbol{k}r}/k_\parallel v_\mathrm{A})^2$ versus $b_k^{1/2}$ for different τ values



$|c\delta\boldsymbol{E}_\perp/v_\mathrm{A}\delta\boldsymbol{B}_\perp|$ versus $b_k^{1/2}$ for different τ values



$|c\delta E_\parallel k_\perp/v_\mathrm{A}\delta\boldsymbol{B}_\perp k_\parallel\tau|$ versus $b_k^{1/2}$ for different τ values

$$\omega_{\boldsymbol{k}r}^2 \simeq k_\parallel^2 v_\mathrm{A}^2 \frac{\sigma_k b_k}{1-\Gamma_k}.$$

$$\left|\frac{c\delta\boldsymbol{E}_\perp}{\delta\boldsymbol{B}_\perp}\right| = v_\mathrm{A}\left[\frac{b_k}{\sigma_k(1-\Gamma_k)}\right]^{1/2}$$

$$\left|\frac{c\delta E_\parallel}{\delta\boldsymbol{B}_\perp}\right| = v_\mathrm{A}\left|\frac{k_\parallel}{k_\perp}\right|\tau\left[\frac{b_k(1-\Gamma_k)}{\sigma_k}\right]^{1/2}$$

Defining $s_k \equiv sign[\omega_k/(k_\parallel v_A)]$ and $\alpha_k \equiv |\omega_k/(k_\parallel v_A)|$,

$$\beta_{k''}^{k'} = (\Gamma_{k'} - \Gamma_{k''}) + (b_{k'} - b_{k''})\frac{s_{k'}}{\alpha_{k'}}\frac{s_{k''}}{\alpha_{k''}}\sigma_{k'}\sigma_{k''} - b_k\frac{s_k}{\alpha_k}\left(\frac{s_{k'}}{\alpha_{k'}} - \frac{s_{k''}}{\alpha_{k''}}\right)\sigma_{k'}\sigma_{k''}$$

$$\begin{aligned}
\beta_{k''}^{k'} &= \sigma_{k'}\sigma_{k''}\left[\frac{\Gamma_{k'} - \Gamma_{k''}}{\sigma_{k'}\sigma_{k''}} + (b_{k'} - b_{k''})\frac{s_{k'}}{\alpha_{k'}}\frac{s_{k''}}{\alpha_{k''}} - b_k\frac{s_k}{\alpha_k}\left(\frac{s_{k'}}{\alpha_{k'}} - \frac{s_{k''}}{\alpha_{k''}}\right)\right] \\
&= \sigma_{k'}\sigma_{k''}\left[\frac{\sigma_{k'}(1-\Gamma_{k''}) - \sigma_{k''}(1-\Gamma_{k'})}{\sigma_{k'}\sigma_{k''}} + (b_{k'} - b_{k''})\frac{s_{k'}}{\alpha_{k'}}\frac{s_{k''}}{\alpha_{k''}} - b_k\frac{s_k}{\alpha_k}\left(\frac{s_{k'}}{\alpha_{k'}} - \frac{s_{k''}}{\alpha_{k''}}\right)\right] \\
&= \sigma_{k'}\sigma_{k''}\left[\left(\frac{b_{k''}}{\alpha_{k''}^2} - \frac{b_{k'}}{\alpha_{k'}^2}\right) + (b_{k'} - b_{k''})\frac{s_{k'}}{\alpha_{k'}}\frac{s_{k''}}{\alpha_{k''}} - b_k\frac{s_k}{\alpha_k}\left(\frac{s_{k'}}{\alpha_{k'}} - \frac{s_{k''}}{\alpha_{k''}}\right)\right] \\
&= \sigma_{k'}\sigma_{k''}\left[\left(\frac{s_{k''}}{\alpha_{k''}} - \frac{s_{k'}}{\alpha_{k'}}\right)\left(b_{k'}\frac{s_{k'}}{\alpha_{k'}} + b_{k''}\frac{s_{k''}}{\alpha_{k''}}\right) + b_k\frac{s_k}{\alpha_k}\left(\frac{s_{k''}}{\alpha_{k''}} - \frac{s_{k'}}{\alpha_{k'}}\right)\right] \\
&= \sigma_{k'}\sigma_{k''}\left(\frac{s_{k''}}{\alpha_{k''}} - \frac{s_{k'}}{\alpha_{k'}}\right)\left(b_k\frac{s_k}{\alpha_k} + b_{k'}\frac{s_{k'}}{\alpha_{k'}} + b_{k''}\frac{s_{k''}}{\alpha_{k''}}\right)
\end{aligned}$$